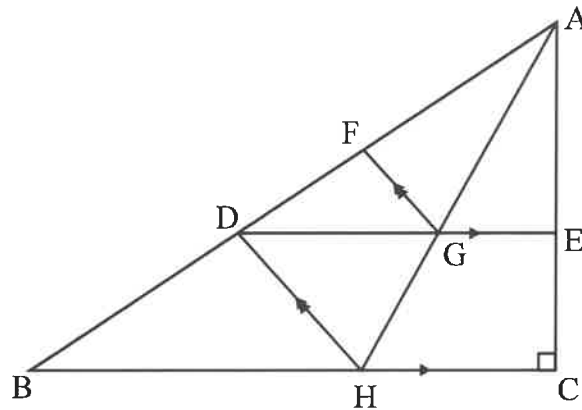


MEMO

QUESTION 1



$$1.1.1 \quad \frac{AG}{GH} = \frac{AE}{EC} = 2 \quad \checkmark A$$

line || side of Δ $\checkmark R$

(2)

$$1.1.2 \quad \Delta ABC \parallel \Delta ADE$$

Given

$$\frac{DE}{BC} = \frac{AE}{AC} = \frac{2}{3} \quad \checkmark A$$

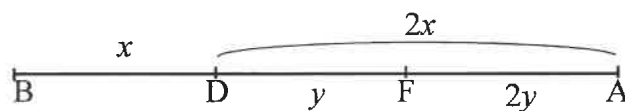
|| Δ s $\checkmark R$

(2)

$$1.1.3 \quad \frac{AF}{FD} = \frac{AG}{GH} = 2$$

 $\checkmark R$ line || side of Δ

$$\frac{AD}{DB} = \frac{AE}{EC} = 2$$

 $\checkmark R$ line || side of Δ Let $BD = x$ and $FD = y$ 

$$2x = 3y$$

 $\checkmark m$

$$x = \frac{2}{3}y$$

$$\frac{AF}{FB} = \frac{2y}{x+y}$$

$$\frac{AF}{FB} = \frac{2y}{\frac{3}{2}y+y}$$

$$\frac{AF}{FB} = 2y \div \frac{5}{2}y$$

$$\frac{AF}{FB} = \frac{4}{5} \quad \checkmark CA$$

(4)

$$1.2 \quad 2x = 16 \text{ cm}$$

$$\therefore x = 8 \text{ cm} \quad \checkmark$$

$$AB = 3(8) \quad \checkmark$$

$$\therefore AB = 24 \text{ cm} \quad \checkmark$$

(3)

$$1.3.1 \quad \frac{\text{Area of } \triangle ADE}{\text{Area of } \triangle ABC} = \left(\frac{AE}{AC}\right)^2 \quad \checkmark$$

$$= \left(\frac{2}{3}\right)^2$$

$$= \frac{4}{9} \quad \checkmark$$

(2)

$$1.3.2 \quad BC^2 = AB^2 - AC^2$$

Pythag

$$BC^2 = 567$$

$$\therefore BC = 9\sqrt{7} \quad \checkmark$$

$$\therefore HC = 3\sqrt{7} \quad \checkmark CA$$

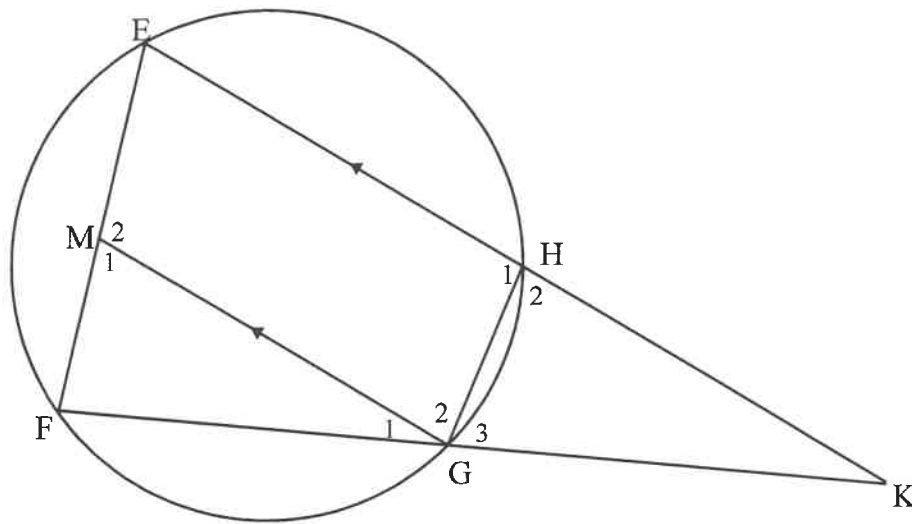
$$\frac{\text{Area of } \triangle ABH}{\text{Area of } \triangle AHC} = \frac{BH}{HC} \quad \checkmark$$

$$= \frac{6\sqrt{7}}{3\sqrt{7}}$$

$$= 2 \quad \checkmark CA$$

(4)

QUESTION 2



2.1 In ΔKGH and ΔKEF :

$\hat{K}$ is common ✓

$\hat{H}_2 = \hat{F}$ S/R Ext $\angle$ of cyclic quad

$\hat{G}_2 = \hat{E}$ S/R Ext $\angle$ of cyclic quad OR 3rd $\angle$ in a Δ

$\therefore \Delta KGH \parallel \Delta KEF$ ✓

(4)

2.2 In ΔKGH and ΔGMF :

$\hat{M}_1 = \hat{E}$ S/R Corr $\angle$'s ; $EK \parallel MG$

$\hat{H}_2 = \hat{F}$ S/R Proven

$\hat{G}_1 = \hat{K}$ S/R 3rd $\angle$ in a Δ

$\therefore \Delta KGH \parallel \Delta GMF$ S/R $\angle \angle \angle$

order of Δ 's

$\therefore \frac{KG}{GM} = \frac{GH}{MF}$ S/R $\parallel \Delta$ s

(5)

2.3 $\frac{KG}{KF} = \frac{EM}{EF}$ S/R line $\parallel$ side of Δ

But $KG = EF$

$\therefore \frac{KG}{KF} = \frac{EM}{KG}$ ✓ Subs in $KG = EF$

$\therefore KG^2 = EM \cdot KF$

(2)

2.4 $\hat{H}_2 = \hat{G}_2$

$\checkmark_R$

Alt $\angle$'s ; EK || MG

$\hat{H}_2 = \hat{F}$

$\checkmark_R$

Proven

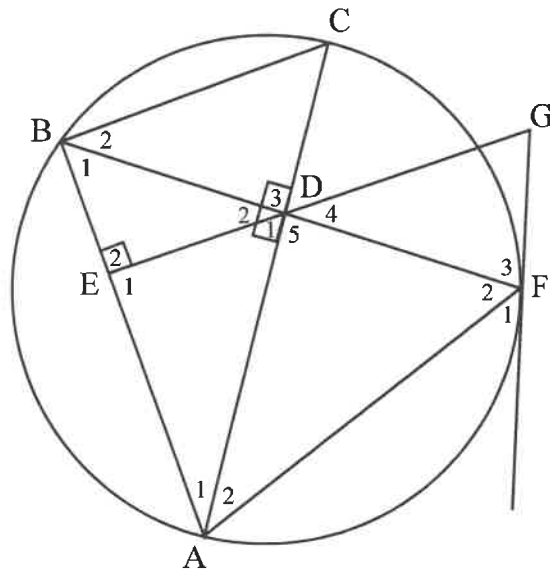
$\therefore \hat{G}_2 = \hat{F} \checkmark_A$

$\therefore$ HG is a tangent

Converse tan chord thrm $\checkmark_R$

(4)

QUESTION 3



3.1 $\angle$ in a semi circle ✓

(1)

3.2 $\triangle BDC$ and $\triangle ADB$ ✓ A

(3)

Line from right $\angle$ vertex $\perp$ hypotenuse ✓ R

3.3.1 $AC = 5x$ ✓

(1)

3.3.2 $\triangle BDC \sim \triangle ADB$

$\sim \Delta$'s to $\triangle ABC$

$$\frac{BD}{AD} = \frac{DC}{BD}$$

S/R

$\sim \Delta$ s

$$BD^2 = AD \cdot DC \quad \checkmark$$

$$BD^2 = 2x \cdot 3x \quad \checkmark \text{ Subs in terms of } x$$

$$BD^2 = 6x^2$$

$$BD = \sqrt{6}x \quad \checkmark$$

(4)

3.4 $\hat{A}_1 = \hat{B}_2$

S/R

$\sim \Delta$ s

$$\hat{B}_2 = \hat{A}_2$$

S/R

$\angle$ s in the same segment

$$\therefore \hat{A}_1 = \hat{A}_2 \quad \checkmark$$

$\therefore AC$ bisects $B\hat{A}F$

(3)

3.5	$\hat{A}_1 = \hat{A}_2 = y$		Proven
	$\hat{F}_3 = 2y$	✓	Tan Chord Thrm
	$\triangle BAD \parallel \triangle BDE$	✓	Line from right $\angle$ vertex $\perp$ hypotenuse
	$\therefore \hat{D}_2 = \hat{A}_1 = y$	✓	$\parallel \Delta$ s
	$\hat{D}_4 = \hat{D}_2 = y$	✓	Vert Opp $\angle$ s
	$\therefore \hat{FGD} = 180 - 3y$	✓ A	$\angle$ s in a Δ ✓

(6)

* Many Alternatives *

[18]